AgriCL-Net

A Lightweight Continual Learning Framework for Plant Disease

Team-6B

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Problem Statement

Plant diseases significantly affect crop yield and quality, while existing plant disease detection systems are mostly limited to fixed disease classes and simple classification. They often lack accurate localization, segmentation, and severity estimation of infected regions and may forget previously learned diseases when new diseases are introduced. Therefore, there is a need for a lightweight and scalable AI-based system that can accurately diagnose plant diseases, estimate their severity, and continuously learn new diseases without losing previously acquired knowledge.

The Proposed Solution

AgriCL-Net unifies detection, segmentation, severity estimation and lifelong learning in one lightweight pipeline

- **Classification** — Swin Transformer identifies the disease present in a plant image.
- **Detection** — DETR localizes the infected regions on the leaf or plant.
- **Segmentation** — SAM2-UNet produces pixel-level masks of the diseased area.
- **Severity Estimation** — Affected-area calculation rates severity as Mild, Moderate, or Severe.
- **Continual Learning** — Replay buffer + knowledge distillation add new diseases without forgetting old ones.
- **Edge-Ready** — Quantization and optimization shrink the model for on-device deployment.

System Workflow

From plant image to severity diagnosis, with continual learning feeding back in



Continual Learning Loop

Replay Buffer (Herding) + Knowledge Distillation let the model learn newly emerging diseases without forgetting previously learned ones — then Model Optimization prepares it for Edge Deployment.

Core Deep Learning Models

Three specialized models, each handling one stage of diagnosis

Swin Transformer

CLASSIFICATION

A hierarchical vision transformer that assigns the disease category to the plant image, forming the first stage of diagnosis.

DETR

DETECTION

An end-to-end transformer-based detector that locates and draws bounding boxes around infected regions of the plant.

SAM2-UNet

SEGMENTATION

Combines SAM2 with a U-Net decoder to produce precise, pixel-level masks of the exact diseased area.

The Continual Learning Engine

How AgriCL-Net learns new diseases without forgetting old ones



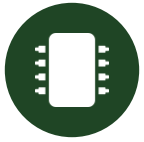
Replay Buffer with Herding

Retains a small, representative set of examples from previously learned diseases, selected via herding, so the model keeps rehearsing what it already knows.



Knowledge Distillation

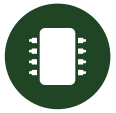
Uses cross-entropy loss and KL divergence to transfer knowledge from the prior model, preserving old disease knowledge while new classes are learned.



Model Optimization & Quantization

Compresses the updated model after each learning cycle, keeping it lightweight enough for deployment on resource-constrained edge devices.

Hardware Requirements



Development & Training

Processor	Intel i5 / Ryzen 5 or better
RAM	Minimum 8 GB, preferably 16 GB
Storage	Minimum 20–50 GB free space
GPU	NVIDIA GPU preferred for local training
Internet	Required for dataset/model downloads & cloud training

Software Requirements

OS & Language

- Windows 10/11
- Linux
- Google Colab environment
- Python 3.11 (recommended)

Frameworks & Libraries

- PyTorch, Torchvision, Transformers
- OpenCV, Pillow
- NumPy, Pandas, Scikit-learn
- Matplotlib, Streamlit

Deep Learning Models

- Swin Transformer — Classification
- DETR — Detection
- SAM2-UNet — Segmentation

Continual Learning

- Replay Buffer + Herding
- Knowledge Distillation
- Cross-Entropy Loss
- KL Divergence

Stakeholders

Who benefits from AgriCL-Net

Farmers

Identify diseases early and understand their severity.

Agricultural Officers

Monitor crop health and advise farmers.

Agricultural Researchers

Study emerging diseases and evaluate AI-based solutions.

Plant Pathologists

Get assisted classification, localization, and severity data.

Agri-Tech Companies

Integrate into farming platforms, apps, drones, and IoT.

Government Bodies

Use aggregated data for large-scale outbreak monitoring.

Relevance to Society

Plant diseases affect crop productivity, farmer income, and food availability — especially where expert access is limited



Early Disease Detection

Farmers can act before disease spreads extensively.



Reduction in Crop Losses

Accurate severity data supports better treatment decisions.



Support for Farmers

Preliminary diagnosis without needing an expert on-site.



Improved Productivity

Better monitoring contributes to healthier crops.



Accessible Technology

Lightweight model runs on affordable edge devices.



Adapts to New Diseases

Continual learning keeps the system useful long-term.

Healthier Crops, Smarter Diagnosis, Continuous Learning

AgriCL-Net brings classification, detection, segmentation, and severity estimation together with a continual-learning core — built to stay accurate as new plant diseases emerge, and light enough to run in the field.

Thank You

AgriCL-Net — Team 6B